Earnings volatility and earnings predictability

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Abstract: Survey evidence indicates widely held managerial beliefs that earnings volatility is negatively related to earnings predictability. In addition, existing research suggests that earnings volatility is determined by economic and accounting factors, and both of these factors reduce earnings predictability. We find that the consideration of earnings volatility brings substantial improvements in the prediction of both short and long-term earnings. Conditioning on volatility information also allows one to identify systematic errors in analyst forecasts, which implies that analysts do not fully understand the implications of earnings volatility for earnings predictability.

Earnings volatility and earnings predictability

1. Introduction

This study investigates the link between earnings volatility and earnings predictability. The motivation for this topic comes from several sources. First, a number of applications require the prediction of earnings (e.g., equity valuation) while our knowledge in this area remains limited, especially for long-run forecasts of earnings. Second, recent survey evidence reveals widely held managerial beliefs that earnings volatility reduces earnings predictability (Graham, Harvey and Rajgopal 2005). This study is a test of the validity and utility of these beliefs. Third, existing findings offer some conjectures about the mechanism that drives the relation between earnings volatility and earnings predictability. We view earnings volatility as arising from two factors, volatility due to economic shocks and volatility due to problems in the accounting determination of income, and both of these factors reduce the predictability of earnings. We present a simple theoretical framework that operationalizes these concepts, and link them to the empirical tests that follow.

The empirical specifications focus on establishing the relation between earnings volatility and short and long-term earnings predictability. To alleviate concerns about a mechanical relation, we use pre-determined measures of volatility to partition the data into volatility quintiles and then use prospective data to estimate earnings predictability. The short-term specifications indicate that low volatility earnings have much higher persistence as compared to high volatility earnings, for a range of persistence of 0.93 vs 0.51 across quintile portfolios. We also find that the strength of the earnings volatility effect exceeds that of several plausible benchmarks, including cash flows volatility, the accrual effect from Sloan (1996) and the extreme-earnings mean-reversion effect from Freeman, Ohlson, and Penman (1982). The results from the long-run tests indicate that earnings volatility has substantial predictive power for up to five years in the future. Earnings with low volatility have remarkably high persistence and R^2 during the entire predictive horizon, while earnings with high volatility show quick reversion to the mean and little reliable predictability. We also document that the results remain qualitatively the same after controlling for two correlated and competing explanations. First, volatile earnings tend to be extreme earnings, and extreme earnings tend to mean-revert faster (e.g., Freeman, Ohlson, Penman 1982), which implies an alternative explanation for why volatile earnings are less persistent. However, the results remain largely unchanged after a control for the level of earnings. Second, volatile earnings are more likely to include transitory items, and since transitory items are less persistent, the documented relation between earnings volatility and earnings predictability could be an artifact of the effect of transitory items. However, the tenor of the results remains the same after controlling for transitory items. A number of additional checks confirms the robustness of the documented relations.

Finally, we investigate whether financial information users understand the implications of earnings volatility for earnings predictability. We use analysts' forecasts as a proxy for sophisticated users' expectation of earnings. We find that conditioning on earnings volatility information allows one to identify large and predictable errors in analysts' forecasts, which suggests that analysts do not fully understand the implications of earnings volatility for future earnings. In quantitative terms, we estimate that analysts impound less than half of the full implications of earnings volatility for earnings predictability.

The remainder of the paper is organized as follows. Section 2 presents the theory of the paper. Section 3 presents the main empirical tests and results, Section 4 contain robustness checks and additional results, and Section 5 presents the results for analyst forecasts tests. Section 6 concludes.

2. Theory and relation to existing research

A number of important applications of accounting data require the prediction of earnings. For example, valuation research and practice typically use projections of earnings to derive estimates of firm and equity value. In fact, existing experience with valuation models like DCF and residual-income suggests that the conceptual differences between valuation models are not that important; what really matters is the extent to which these models help in the empirical projection of future fundamentals, usually based on projected future earnings. A related application is the use of accounting data to derive and possibly improve on analysts' earnings forecasts. On the one hand, analysts are continually looking for new ways to more accurately predict earnings. On the other hand, investors are interested in ways in which they can identify biases in analysts' forecasts and improve on the accuracy of existing forecasts. On a more general level, such applications and needs are related to the rise of "fundamental analysis" research in accounting, where fundamental analysis can be defined as identifying ways to use accounting data to produce superior forecasts of earnings (e.g., Nissim and Penman 2001).

Given these needs, our knowledge about the predictability of earnings is limited, especially for long-term predictability. There are a number of useful models and results for one-year ahead forecasts, e.g., mean reversion, the Foster (1977) model of quarterly earnings, the accrual effect due to Sloan (1996), and the fundamental analysis signals due to Lev and Thiagarajan (1993) and investigated in Abarbanell and Bushee (1997). In contrast, there are few useful long-term results. This dearth of results seems unsatisfactory because some key applications (e.g., equity valuation) require long-term forecasts of earnings, and it is the accuracy of the forecasts which drives the success of these applications. In fact, the typical projection of long-term earnings relies only on mean reversion, and the only real differences between various empirical specifications are about what the eventual "steady-state" mean is, and about the rate of fading to that mean. Thus, in spite of some contributions in this area due to Finger (1994) and Penman and Zhang (2002), our knowledge about the long-term predictability of earnings remains rudimentary.

We aim to enhance the knowledge in this area by investigating the relation between earnings volatility and earnings predictability. Our motivation stems from several sources which suggest that earnings volatility captures aspects of the determination of earnings which are related to the predictability of earnings. First, recent survey evidence offers strong motivation for the link between earnings volatility and earnings predictability. Graham, Harvey, and Rajgopal (2005) survey 401 financial executives to determine the key factors that drive decisions related to reported earnings and find a pronounced aversion to earnings volatility (97 percent of respondents express a preference for smooth earnings). In exploring the reasons for this finding, the authors find that executives abhor volatility because it is thought to reduce the predictability of earnings (80 percent of respondents express this belief). Thus, our investigation is a test of widely held managerial beliefs that earnings volatility is negatively related to earnings predictability. The investigation also helps to map out the specific content of this relation. The survey evidence leaves little doubt that executives believe that more volatile earnings are less predictable. However, it is less clear what the executives have in mind by “predictable”, and it is possible that the executives view these concepts as semantic or tautological opposites, so to them the relation obtains by construction. The analysis later provides a framework for defining and operationalizing the difference between these concepts, which is then reflected in the empirical tests.

Although the survey evidence does not provide clues about the specific mechanism relating earnings volatility to earnings predictability, we posit that this relation is due to both economic and accounting factors. On the more obvious level, earnings volatility captures the effects of real and unavoidable economic volatility. Intuitively, firms operating in environments subject to large economic shocks are likely to have both more volatile earnings and less predictable earnings. Although the theory about this link seems straightforward, there is little empirical evidence about it. To our knowledge, only Lipe (1990) explores the relation between economic volatility and earnings predictability in a short-horizon setting. However, this relation is a side issue for Lipe (1990) and from his evidence it is difficult to gauge the economic and long-term importance of this relation.

On a more subtle level, the volatility of reported earnings also reflects important aspects of the accounting determination of income, which also provide a link to earnings predictability. One such aspect is the quality of matching of expenses to revenues, as modeled in Dichev and Tang (2008). The basic idea in Dichev and Tang is that poor matching acts as noise in the economic relation between revenues and expenses, and thus the volatility of reported earnings increases in poor matching. Poor matching is also associated with poor earnings predictability because the matching noise in reported earnings obscures the underlying economic relation that governs the evolution of earnings over successive periods. Thus, the joint effect of poor matching on earnings volatility and earnings predictability suggests another link between these two variables. The quality of accruals effect in Dechow and Dichev (2002) is another aspect of the determination of earnings which provides an accounting link between earnings volatility and earnings predictability. Dechow and Dichev argue that many accruals estimate future cash flows, and therefore large magnitudes of estimation errors in accruals signal lower quality of earnings and lower predictability of earnings. Since estimation errors are likely to be more serious in volatile environments, this suggests a negative relation between earnings volatility and earnings predictability.

It is also possible that the link between earnings volatility and earnings predictability reflects other factors, e.g., earnings smoothing behavior, where managers smooth earnings to provide a more predictable measure of firm performance. In any case, for our purposes the distinction among plausible causes is not that important because the focus is on investigating for the existence and the economic importance of the relation between earnings volatility and earnings predictability, rather than on its explanations. Later, we provide evidence on the relative role of common economic and accounting factors in the documented relations.

We start our investigation with some theoretical considerations. The goal is to provide a simple framework that formalizes the preceding motivations and link them to the empirical analysis that follows. Our analysis of the relation between earnings volatility and earnings predictability relies on commonly used autoregressive regressions of current on one-year lagged earnings.

$$E_t = \alpha + \beta * E_{t-1} + \varepsilon \qquad (1)$$

Taking the variance of both sides yields:

$$Var(E_t) = \beta^2 * Var(E_{t-1}) + Var(\varepsilon) \qquad (2)$$

Assuming that the variance of earnings is stationary over time[1], and re-arranging obtains:

$$Var(\varepsilon) = Var(E) * (1-\beta^2) \qquad (3)$$

Expression (3) is a useful summary of the key variables and relations of our study. Var(E) is our proxy for volatility of earnings. Var(ε) is our (inverse) proxy for "earnings predictability", because the variance of the error term captures the variation in earnings remaining after accounting for the effect of the autoregressive coefficient, β.

[1] Existing research indicates that the volatility of earnings has approximately doubled over the last 40 years, see Givoly and Hayn (2000) and Dichev and Tang (2008). However, the stationarity argument holds reasonably well for the one-year horizon used here.

Equation (3) is also a useful guide to the mechanism of the link between earnings volatility and earnings predictability, revealing a two-fold relation. First, holding earnings persistence constant, earnings volatility is inversely related to earnings predictability. Second, this negative relation is likely strengthened through the effect of the persistence coefficient because, as discussed above, there are reasons to believe that β itself is negatively related to volatility of earnings. For example, economic or accounting noise in earnings is likely to both increase the volatility of earnings and decrease the persistence of earnings. Note that there is no statistical reason to expect a relation between Var(E) and β. The volatility of the autoregressive variable can be high or low, and it has no necessary relation to persistence. To illustrate this point, consider the behavior of stock prices. Under the maintained assumption of market efficiency, stock prices are random walks, and thus, the persistence of the autoregressive relation in prices is always one, regardless of the volatility of the stock.

To formally examine the mechanism of the link between earnings volatility and earnings predictability, we take the total derivative of the variance of the error term with respect to earnings volatility. Using expression (3), and denoting total (partial) derivative as d (δ), yields:

$$d[Var(\varepsilon)]/dVar(E) = (1-\beta^2) - 2*Var(E)*(\delta\beta/\delta Var(E)) \qquad (4)$$

The first term in Equation (4) suggests that the strength of the direct relation between earnings volatility and earnings predictability is determined by earnings persistence, where higher persistence signifies more predictable earnings. The second term in Equation (4) represents the second link between earnings volatility and earnings predictability through the effect of earnings volatility on earnings persistence. More specifically, the hypothesized negative effect of earnings volatility on earnings persistence should reinforce the base negative relation between earnings volatility and earnings predictability.

Note that the notion of predictability captured in Var(ε) is "absolute" predictability, unadjusted for volatility in the earnings environment. If one is interested in "relative" predictability, a natural scalar for Var(ε) is Var(E). Taking (3), dividing it by Var(E), and re-arranging leads to:

$$1 - \mathrm{Var}(\varepsilon)/\mathrm{Var}(E) = \beta^2 \tag{5}$$

Expression (5) simply says that relative predictability is the R^2 of the regression, which is equal to the squared persistence coefficient. Thus, identifying the relation between earnings volatility and earnings persistence is a key to our investigation of both absolute and relative earnings predictability.

We use the insights from this framework in the empirical tests on two dimensions. First, we map out the economic importance of the conjectured negative relation between earnings volatility and short-term and long-term earnings persistence. Second, we investigate whether and how the use of earnings volatility information leads to appreciable gains in earnings predictability. Specifically, in out-of-sample tests we investigate whether conditioning on earnings volatility information leads to lower prediction errors as compared to other popular models of earnings prediction. In further tests, we check whether analyst forecasts impound the implications of earnings volatility information by investigating whether conditioning on volatility information allows one to identify systematic errors in analyst forecasts.

The exploration of the link between earnings volatility and earnings predictability seems warranted because we are not aware of other studies that provide a direct and comprehensive investigation of this relation. Minton, Schrand and Walther (2002) use an underinvestment motivation and find that firms with high cash flow volatility have lower *levels* of future cash flows and earnings. Note that this is different from our prediction that high earnings volatility results in lower *persistence* and predictability in future earnings. Thus, the Minton, Schrand and

Walther effect and our effect are complementary, and in fact we illustrate how to combine these two effects later in our study.

3. Main empirical tests

3.1 Sample selection, descriptive statistics, and test specification

Table 1, Panel A summarizes the sample selection. Our sample is obtained from the Compustat annual industrial and research files over 1984 to 2004. We restrict the sample to this period because we need cash flow statement data for the accurate estimation of accruals and cash flows (Collins and Hribar 2002). Cash flow statements become widely available since 1988, and we use the preceding years over 1984 to 1988 to calculate the volatility of earnings. The sample is restricted to firm-years with complete data for assets (Compustat item 6), earnings (item 123), cash flow from operations (item 308), and preceding four years of earnings and cash flows from operations. Accruals are estimated by taking the difference between earnings and cash flows from operations. Earnings, accruals and cash flow from operations (CFO) are deflated using average assets.[2] Earnings volatility is calculated by taking the standard deviation of the deflated earnings for the most recent five years (the tenor of the results remains the same if the earnings volatility variable is based on the five years of earnings *preceding* the current year). Cash flow volatility is calculated by taking the standard deviation of the deflated cash flows for the most recent five years. To avoid the influence of extreme observations, we truncate the top and bottom 1 percent of earnings, accruals and cash flows from operations. In addition to these fairly common sample selection criteria, we impose two additional requirements. First, we limit the sample to economically substantial firms, defined as a minimum of $100 million in assets. Our

[2] Results are similar using an undeflated (EPS) specification. Results using a price deflator have the same tenor but are substantially weaker than those using an asset deflator, possibly because price itself is s function of earnings rather than being a neutral deflator.

concern is that small firms tend to be economically negligible but statistically influential (because of extreme realizations). Second, we limit the sample to 12/31 fiscal year-end firms to simplify the tests and the interpretation of the results. Later, we provide some evidence on the effect of these restrictions on our results. After all requirements, the final sample includes 22,113 firm-years over 1988 to 2004.

Descriptive statistics for the full sample are presented in Table 1, Panel B. The results are in line with much other research that explores similar variables and time period. Cash flow from operations is typically higher than earnings (mean of 8.5 percent vs. 3.1 percent), and accruals are negative (mean of -5.5 percent). Firm-specific volatility of scaled earnings has a mean of 4.0 percent and a large standard deviation of 15.7 percent, indicating large differences in earnings volatility across firms. The descriptive statistics for volatility of earnings also reveal that this variable has a highly non-normal distribution, bounded at 0 on the left and heavily right skewed. To address such non-linearities and aiming for a more robust estimation in general, much of the subsequent analysis relies on quintile portfolios formed on conditioning variables, mainly volatility of earnings. The portfolio-based analysis also provides an immediate and clear reflection of the economic importance of the results.

Firms in the full sample are randomly assigned into two sub-samples. We use the first sub-sample (observations = 11,061) for a comprehensive exploratory analysis of the predictive power of earnings volatility for earnings predictability, while the second sub-sample (observations = 11,052) is used to perform out-of-sample tests of forecasting accuracy.

3.2 Results for one-year predictive horizons

Table 2 presents the persistence coefficients and R^2 of regressions of one-year ahead earnings on current earnings. As discussed above, these results provide evidence about the economic and statistical significance of the hypothesized negative relation between earnings volatility and earnings persistence. While the persistence coefficients and the R^2 are clearly related in these regressions, they also differ because the conditioning variables often provide for systematic differences between the variability of current and future earnings. Baseline results for the full sample in Panel A reveal a persistence coefficient of 0.65 and R^2 of 0.39, in line with existing results for this specification.

Panel B of Table 2 presents the results for quintiles formed on volatility of earnings. An examination of Panel B reveals that there is a strong and monotonic relation between volatility of earnings and earnings persistence. The persistence coefficient declines from 0.93 in quintile 1 to 0.51 in quintile 5 and the adjusted R^2 declines from 0.70 in quintile 1 to 0.30 in quintile 5. These declines seem large in absolute magnitude and suggest that conditioning on earnings volatility is economically important. Panel B also provides tests of the statistical significance of these differences, specifically the differences for persistence and R^2 between quintiles 1 and 5. The test for difference in persistence is a simple t-test from a regression that combines quintile 1 and 5 observations, with dummy intercept and slope variables for quintile 5 observations. Testing for difference in R^2 is more problematic because it involves comparing R^2 across two essentially different regressions. Although the dependent variable looks the same (future earnings), traditional tests like the Vuong test are inappropriate because the variation of the dependent variable is quite different across earnings volatility quintiles. Instead, we use a bootstrap test based on simulating the empirical distribution of the test statistic, assuming that the null is true (Noreen 1989). In this case, the null hypothesis is that earnings volatility is unrelated to earnings predictability, and the test statistic is the difference in adjusted R^2 between earnings volatility quintiles 1 and 5. We simulate the empirical distribution under the null by randomly splitting the full sample (11,061 observations) into pseudo-earnings volatility quintiles. Then, we run the earnings persistence regression within pseudo quintiles 1 and 5, and obtain a difference in R^2 between the two quintiles. This difference is one observation from the simulated distribution under the null. We repeat this procedure 1,000 times, yielding a 1,000-observation empirical distribution of R^2 differences under the null. The formal statistical test is based on comparing the actual observed difference in R^2 against the simulated distribution of differences. For Panel B, the statistical tests indicate that the differences in persistence and R^2 between quintile 1 and 5 for earnings volatility are highly significant (both $p < 0.001$).

In Panels C and D of Table 2 we provide results about earnings persistence by conditioning on the level of accruals and level of earnings. These other results serve two purposes. First, they provide a benchmark for the economic magnitude of the earnings volatility results. The level of accruals variable is motivated by Sloan (1996), who shows that it is a powerful determinant of future earnings persistence. The level of earnings variable is motivated by much existing research, which documents that extreme earnings tend to mean-revert faster, i.e., level of earnings is a determinant of earnings persistence (Freeman, Ohlson, Penman 1982). Second, the results in Panels C and D provide evidence on whether the earnings volatility predictive effect is incremental to those of existing effects since volatility of earnings is likely correlated with both level of accruals and level of earnings. To make the results comparable across panels, we define all variables in a way that yields highest earnings persistence in quintile 1 and lowest persistence in quintile 5.

Panel C illustrates the Sloan (1996) result in our sample by conditioning on level of absolute accruals. Since Sloan (1996) shows that extreme accruals are less persistent, we expect that the persistence of earnings is lower in quintile 5.[3] Indeed, the persistence of earnings for

[3] There are some differences between our specification of level of accruals and that of Sloan (1996). First, Sloan uses a balance sheet-based derivation of accruals, while we use the more accurate cash flow-based method. Second,

quintile 5 is about 0.55, much lower than the 0.78 to 0.87 range for the rest of the accrual quintiles. R^2 for quintile 5 is also lower, and both the persistence and the R^2 differences across extreme quintiles are statistically significant. Turning to a comparison of the results across Panels B and C, we find that the decline in persistence across earnings volatility quintiles (0.43) is moderately higher than the decline for the accrual quintiles (0.33). The same pattern of results is observed for R^2 but the decline in R^2 across earnings volatility quintiles (0.41) is much larger than the corresponding decline for the accrual quintiles (0.12).

We also perform bootstrap tests for the statistical significance of the across-quintiles differences *across panels*, e.g., is the across-quintiles difference in persistence in Panel B (0.43) greater than the across-quintile difference in Panel C (0.33). Specifically, the tests construct random pseudo-earnings volatility quintiles, run regressions within the quintiles, and obtain a difference in persistence and R^2 across quintiles. Then another pseudo-level of accruals simulation is run and across-the-panels differences in persistence and R^2 are produced.[4] This procedure is repeated 1,000 times, and the actual differences are compared to the simulated distribution of differences. The results indicate that the difference in persistence ranges across Panels B and C (0.43 vs. 0.33) has a p-value of 0.009 and the difference in R^2 ranges (0.41 vs. 0.13) has a p-value < 0.001. Summarizing, a comparison of the results across Panels B and C suggests that earnings volatility dominates level of accruals in terms of predictive power.[5]

Sloan uses raw level of accruals, while we use absolute level of accruals because we want a measure of quality of earnings that is monotonic in level of accruals. Of course, there are sample and time-period differences as well.

[4] Note that this specification assumes that the sorts across panels are independent, while earnings volatility and level of accruals are empirically correlated. However, since the test is on across-panel *differences*, and the two sorting variables have a *positive* correlation, the resulting null distribution under the assumption of independence is *conservative* (if one simulates a positive correlation between the sorting variables for producing the null, the resulting differences will have a tighter distribution). The same considerations apply for all other across-panel bootstrap tests because all other variables are also positively correlated with volatility of earnings, e.g., extremeness of current earnings, volatility of cash flows from operations.

[5] We have also performed a number of additional tests that explore the incremental and joint explanatory power of earnings volatility and level of accruals by using the two-pass sorts used in Dechow and Dichev (2002) and 5X5 sorts on both variables. The results reveal that the explanatory power of these two variables is largely incremental to

Panel D presents the results for level of earnings quintiles. Earnings are first sorted on their magnitude into deciles 1 to 10, and then the deciles are combined into quintiles, where deciles 1 and 10 form quintile 5, deciles 2 and 9 form quintile 4, and so on. Since quintile 5 comprises the most extreme earnings, we expect it to have the least persistent earnings; the opposite holds for quintile 1. Indeed, an inspection of Panel D reveals that the persistence of earnings decreases across quintiles, from 0.79 in quintile 1 to 0.62 in quintile 5. However, the resulting range of 0.18 is much smaller than the corresponding range of 0.43 for earnings volatility, and this difference has a p-value < 0.001 in bootstrap tests of significance. Thus, these results suggest that the earnings volatility effect cannot be subsumed by the level of earnings effect in earnings predictability. We provide further and more specific evidence about the incremental effect of these two variables in the section on long-run earnings predictability. Also, note the pattern in R^2 goes in the opposite direction, decreasing from 0.54 in quintile 5 to 0.03 in quintile 1, which at first seems surprising. Further reflection suggest that this is to be expected, given that $R^2 = \beta^2 * Var(E_t)/Var(E_{t+1})$ and that by construction the variance of the independent variable is much more limited for the lower quintiles in Panel D.

Panel E presents results for one more conditioning variable, volatility of cash flows, which serves as a proxy for economic volatility. Recall that Section 2 suggests that one advantage of the earnings volatility variable is that it combines the explanatory power of both economic volatility and accounting problems-based volatility with respect to earnings predictability. If this conjecture is true, we expect that earnings volatility has higher explanatory power than cash flow volatility with respect to earnings predictability. An examination of Panel E reveals that volatility of cash flows provides a good ranking on earnings predictability, with

each other and that a joint consideration of both variables yields better results that the consideration of either one alone.

range in persistence of 0.18 and range in R^2 of 0.19. However, the ranges in persistence and R^2 for the earnings volatility variable in Panel B are more than double those in Panel E and the across-panel differences in persistence and R^2 have highly significant p-values. Thus, the results in Panel E suggest that earnings volatility dominates cash flow volatility with respect to earnings predictability. Having in mind that the volatility of cash flows is similar in magnitude to the volatility of earnings (see Table 1, Panel B), this result implies that the volatility in earnings due to the accounting process is important in relation to earnings predictability. In additional untabulated tests, we use sales volatility as another proxy for economic volatility and find that the results for sales volatility are similar to those for cash flow volatility.

3.3 Results for five-year predictive horizons

Table 3 presents results for five-year ahead prediction of earnings, conditional on earnings volatility. Benchmark results for the full sample are presented in Panel A, comprising unconditional regressions of various-horizon future earnings on current earnings. An examination of Panel A reveals that the predictive power of earnings quickly deteriorates for longer prediction horizons, consistent with existing results. The persistence coefficient on earnings drops from 0.65 in year t+1 to 0.38 in year t+5, and R^2 drops from 0.40 in year t+1 to 0.11 in year t+5.

In investigating the effect of earnings volatility, for parsimony we focus the presentation on the extreme quintiles. Panel B in Table 3 presents the results for firm-years in the highest quintile of earnings volatility and Panel C presents the results for the lowest quintile of earnings volatility. Even a cursory examination of these two panels reveals dramatic differences in the long-run predictive characteristics of the underlying samples. High-volatility firm results in Panel B show a quick deterioration of persistence (0.51 to 0.18) and R^2 (0.30 to 0.03) over the five-year predictive horizon, where at all time horizons the numbers in Panel B are lower than those in Panel A. In contrast, the results for low-volatility firms in Panel C reveal a robust predictive power over the entire five-year horizon. The persistence coefficient is high in year t+1 (0.93) and deteriorates only modestly to 0.81 in year t+5. The erosion in R^2 is more substantial (0.70 to 0.32) but in terms of absolute magnitude even for year t+5 one retains a considerable amount of confidence in the prediction of earnings. In fact, a literal reading of these numbers implies that it is easier to predict earnings *five* years ahead for low volatility firms than to predict earnings *one* year ahead for high volatility or even all firms. The combined pattern of these results suggests that earnings volatility has a remarkable differentiating power in the long-run prediction of earnings.[6]

The three panels of Figure 1 present a graphical view of the results in Panels A to C in Table 3. The graphs use a consistent scale to plot the evolution of median profitability over the next five years conditional on quintile portfolios formed on a ranking of current profitability for the full sample (Figure 1a), the high earnings volatility quintile (Figure 1b), and low earnings volatility quintile (Figure 1c). The benchmark results in the first graph reveal the expected mean reversion, where the current-earnings portfolio range of profitability of 0.13 is reduced to about half by the end of the five-year horizon. Consistent with impressions from the statistics in Table 3, the second graph reveals a much faster mean reversion for the high earnings volatility firms – the range in median profitability is reduced from 0.25 in year t to about a fifth of that amount in year t+5. In contrast, there are no visible signs of mean reversion in the third graph. The range

[6] Since Table 2 identifies substantial differences in one-year predictive power across earnings volatility quintiles and the specifications in Table 3 are autoregressive, the long-run differences in Table 3 are partly to be expected. The reason is that, even with no other relations, the β coefficient in a one-year ahead specification will appear as a β^5 coefficient in the five-year ahead specification. However, an examination of the results suggests that there are longer-horizon relations beyond the simple compounding of the first-order autocorrelation in earnings. For example, the first-year persistence coefficient for high volatility earnings is 0.507, and $0.507^5 = 0.033$, which is substantially smaller than the actual 5th-year coefficient of 0.177. The corresponding numbers for the low volatility quintiles are $0.934^5 = 0.711$, which is smaller than the actual coefficient of 0.805 for the 5th year.

in median profitability of about 0.06 is maintained virtually unchanged until year t+5. In addition, the lines for all quintiles look nearly perfectly straight, never intersecting or even reducing the dividing distances between them. To our knowledge, this is the first demonstration of a large-sample setting, which allows for such a clean, simple, and long-lasting differentiation in profitability.

The graphs in Figure 1 also illustrate the confounding effect of the previously discussed relation between earnings volatility and level of earnings. Firms with high volatility of earnings have a larger dispersion in the level of current earnings, so they are expected to have a faster mean reversion as well. Thus, in calibrating the relation between earnings volatility and earnings predictability it is important to control for level of current earnings. We control for the level of current earnings by using a two-pass sorting procedure. Specifically, each year observations are first sorted into 20 portfolios based on the magnitude of their current earnings. Then, within each of these 20 portfolios, observations are further sorted into earnings volatility quintiles. Combining the highest volatility quintiles from portfolios 1 to 4 produces Quintile 1 (low earnings magnitude) for our high earnings volatility subsample, combining the highest volatility quintiles from portfolios 5 to 8 produces Quintile 2, and so on. We repeat the same procedure to derive the quintiles for the low earnings volatility subsample.

The results from this two-pass sorting are presented in the three graphs in Figure 2. The first graph in Figure 2 presents the benchmark results for the full sample, and is identical to the first graph in Figure 1, except for a different scaling. The second graph presents the results for the high volatility subsample, and the third graph present the results for the low volatility subsample, where both should have similar dispersion of current earnings. A comparison of the second and third graph with the first graph reveals that the two-pass procedure is successful in controlling for the dispersion of current earnings. Median current earnings for Quintiles 2 to 5 are nearly identical across graphs, while the control is less successful but seems satisfactory for Quintile 1. An examination of the rest of the graphs reveals clear evidence of differential mean-reversions across graphs, where higher volatility firms revert faster and stronger. While the range in current median earnings is 0.13 and deteriorates to 0.06 in year t+5 for the full sample of firms, the corresponding numbers are 0.15 to 0.02 for the high volatility subsample, and 0.11 to 0.07 for the low volatility subsample. To complete the analysis with control for the dispersion of current earnings, Panels D and E in Table 3 include regression results for the high and low volatility subsamples, controlling for the level of current earnings. The results in Panels D and E largely agree with the corresponding no-control results in Panels B and C in Table 3 - high volatility firms have considerably lower predictability of long-run future earnings. In fact, the control for the dispersion of current earnings seems to have only a marginal effect on the magnitude of the results. Thus, the volatility of earnings effect seems to be largely incremental to the level of earnings effect in the predictability of earnings.

For a cleaner and more compact version of the long-run results and for formal statistical tests, Table 4 provides another specification of the relation between earnings volatility and long-run earnings predictability. In this case, the sum of subsequent five-year earnings is regressed on current earnings. Thus, the coefficient on the independent variable can be interpreted as the five-year sum of the yearly persistence coefficients, while the R^2 provides an aggregate measure of explanatory power over the five-year horizon. The statistical tests in Table 4 are similar to those in the one-year specifications in Table 2, with t-tests of difference in persistence and bootstrap tests for differences in R^2. For clarity of exposition, the results of Panels A through C in Table 4 correspond to the results in Panels A through C in Table 3. An inspection of the results in Table 4 indicates that both the tenor and the magnitude of the results are much the same as in Table 3. The persistence and R^2 for the high volatility sample (1.37 and 0.17) are substantially lower than those for the benchmark full sample (2.36 and 0.31) and much lower than those for the low volatility sample (4.21 and 0.63). The persistence and R^2 differences between the high and low volatility samples are both significant at the 0.001 level.[7]

In Figure 3 we present a portfolio specification that provides an intuitive feel for the economic importance of our long-run earnings volatility results. Figure 3 presents medians of five-year future earnings for two portfolios constructed to control for the level of current profitability and maximize future earnings differences based on earnings volatility information. Specifically, the full sample of firm-years is first sorted yearly into 20 portfolios based on the magnitude of current earnings, which are then further sorted into earnings volatility quintiles. Then we combine the four subportfolios, which have the highest current profitability but fall in the lowest quintile of earnings volatility (high earnings/low volatility) and compare it with the four subportfolios, which have the highest current profitability and fall in the highest quintile of earnings volatility (low earnings/high volatility). The motivation is that high earnings are expected to mean-revert but the mean-reversion will be minimal for the low volatility of earnings portfolio and large for the high volatility portfolio, which will produce predictable differences in future earnings.

Note that the evidence in Figure 3 is limited to high earnings firms because we want a setting which provides a sharp *directional* prediction about future earnings. Medium-profitability firms are not included because their earnings are expected to stay largely the same. Low earnings firms are also excluded because our effect and the Minton, Schrand, and Walther (2002) results contradict and likely cancel each other in the domain of low earnings. Based on our results, firms with low profitability/high volatility should mean-revert faster, which means

[7] Additional tests corresponding to Panels D and E in Table 3 confirm that all results hold after control for level of current earnings.

they should have higher future profitability. But based on Minton, Schrand and Walther (2002), firms with high volatility should have lower future earnings. In contrast, these two effects reinforce each other in the domain of high earnings. Firms with high earnings/high volatility should have a substantial decline in future earnings both because high volatility earnings are less persistent and because on average high volatility firms have lower future earnings.

An examination of Figure 3 reveals that the two-pass procedure controls for current profitability nearly perfectly; thus, all differences in future profitability can be described as gains from using earnings volatility information. For future earnings, the graph reveals a sharp and immediate diversion in profitability, which starts in year t+1 and continues unabated until year t+5. As conjectured, future earnings of the high earnings/low volatility portfolio decrease but only slightly, while the earnings for the other portfolio sharply decrease from current levels. The magnitude of the resulting difference is about 3 percent and it persists over the whole five-year horizon, which seems economically large.

The portfolio differences in Figure 3 also seem large compared to existing results, e.g., a comparable graph in Penman and Zhang (2002) shows that the conservatism “hidden reserves” effect produces five-year differences in profitability on the magnitude of 1.5 to 2 percent, which proves to be economically substantial in their setting. Of course, there are many differences in the motivation, sample, variable definition, and portfolio selection procedure between our study and Penman and Zhang (2002), which suggest that comparisons have to be made with caution. Nevertheless, the combined impression from the results is that a consideration of earnings volatility brings substantial improvements in long-run earnings predictability. Later in the paper, we use the portfolios in Figure 3 to investigate whether financial analysts understand and impound the implications of earnings volatility for earnings predictability.

4. Robustness checks and additional results

First, we investigate the effect of transitory items on our results. Transitory items both increase the volatility of earnings and decrease earnings predictability, so potentially they could be a large determinant of the effects documented in this study (see the Appendix for a more rigorous exposition of this point). Note that the removal of these items and the interpretation of these results have to be done with caution because transitory items like restructurings and asset write-offs are a prime manifestation of both the economic and the poor-accounting aspects of earnings volatility addressed in this paper. Thus, this is really more of a test for whether the documented results are driven by a small subset of observations. In operational terms, we check for the effect of transitory items by repeating the main tests after the elimination of all firm-year observations where the sum of special items (Compustat item 17) and non-operating income/expense excluding interest income (item 190) exceeds 5 percent of total assets. The resulting sample has 9,652 observations, about 13 percent less than the original sample of 11,061 observations. Consistent with intuition and existing evidence, all predictability results improve after the elimination of transitory items, where the effect is minimal for the low-volatility quintiles and much more pronounced for the high-volatility quintiles. However, the tenor and even the magnitude of the results remain substantially the same. For example, the persistence for the lowest volatility quintile declines from 0.92 to 0.81 over the five-year horizon (and R^2 declines from 0.69 to 0.30), while the corresponding numbers for the highest volatility quintile are 0.65 to 0.28 (and R^2 from 0.34 to 0.04). We also repeat the main tests by including year dummies to control for the temporal rise in the importance of special items (e.g., Collins, Maydew, and Weiss 1997). The results remain largely the same. For example, the across-quintile difference in the persistence of earnings for firms with the lowest earnings volatility and firms with the highest earnings volatility is 0.432 (p-value = 0.001), virtually identical to the across-quintile difference of 0.427 as presented in Table 2, Panel B. Thus, the documented strong relation between earnings volatility and earnings predictability is rooted in the properties of the full sample and is not limited to the effect of transitory items or the rising frequency of special items over time.

We also investigate survivorship biases. Ex ante, our results seem less prone to such biases because our sample is limited to economically substantial firms. Nevertheless, Table 3 indicates that the survivorship issue remains valid because there is a large drop-off in the number of available observations over the five-year horizon. To provide some evidence on this issue, we repeat all major tests on a constant sample of 4,032 observations that have at least five years of earnings into the future (thus, the constant sample has a look-ahead bias).[8] Both the short-term and the long-term results for the constant sample are very similar to those presented in the paper.

Another set of results is motivated by the observation that our volatility measures are historical and assume independence over time, while earnings are autocorrelated and differ in persistence. *Ceteris paribus*, firms with high past persistence tend to have low past variance, so the documented effect could be simply the carryover of past persistence into the future. In contrast, the motivations of the study are more along the lines of economic and accounting noise, which are more naturally captured in the residual from past persistence. To offer some evidence on the relative roles of these factors, we employ two specifications. The first specification is to sort on the variance of the residuals from an AR(1) model over the last five years (instead of the variance of raw earnings over the same period); by construction these residuals capture the variation in earnings after the effect of persistence is taken out. An examination of the results in

[8] Since the number of observations drops off more steeply for high volatility firms, we use two alternatives for the portfolio assignments of the constant sample. One specification is based on the portfolio assignments in the original sample (so, the resulting portfolios have differing numbers of observations), while the other specification is based on quintile assignment within the constant sample (the resulting portfolios have the same number of observations). The results for these two specifications are similar.

Table 5, Panel A reveals that the extreme-quintile spread in the persistence coefficient is 0.318 and the spread in R-squared is 0.321, both statistically significant. A comparison with the corresponding benchmark results in Table 2, Panel B (0.427 and 0.408) reveals that the effect of residuals dominates that of persistence. The second specification estimates historical persistence directly and then examines whether the effect of historical volatility is incremental to the effect of historical persistence in a 5X5 sort in Table 5, Panel B.[9] The results indicate that past persistence has little effect on forward-looking persistence after controlling for volatility of earnings. In contrast, there are large spreads in forward-looking persistence across volatility quintiles after controlling for past persistence. Thus, both specifications indicate that the role of past persistence in explaining the documented results is small.

We also explore the effect of cross-sectional dependence in earnings on our tests of significance. Cross-sectional dependence arises because of economy-wide, industry, and other systematic factors in earnings, and could result in understated standard errors and inflated levels of significance. Since most of our tests rely on bootstrap methods of assessing significance, we limit our robustness checks to only the relevant subset of OLS results. We use Fama-MacBeth regressions, which rely on time-series independence to provide tests of significance, and are the most common remedy for cross-sectional dependence. Note that Fama-MacBeth tests are a rather conservative method to estimate statistical significance in our sample because the time-series is relatively short (only 16 years). The results confirm that the documented relations are significant, e.g., the 0.427 range of persistence coefficients across earnings volatility quintiles in Table 2, Panel B has a p-value of 0.001.

[9] The 5X5 sort in Panel B uses independent sorting on the two variables to avoid handicapping one variable vs. the other. In any case, the Spearman correlation between historical persistence and historical volatility is -0.015, (p-value = 0.124), which suggests that the results under sequential sorting schemes will be similar.

We also provide evidence on the relative role of time-series vs. cross-sectional earnings predictability effects on our results.[10] Note that our motivation largely relies on economic and accounting arguments which suggest a relation between firm-level volatility and persistence in earnings. Similar to other existing studies, though, the main regressions are run on panel data and thus, the estimated coefficients are a function of both firm-level autoregressive persistence and variation in mean profitability across firms (see the Appendix for a more rigorous exposition of this point). Since our paper aims for enhancing practical earnings prediction, we are interested in total earnings persistence and total predictive ability, regardless of whether it comes from the autoregressive or the cross-sectional aspect of the regression. Thus, the main results in the paper rely on the estimated coefficients of persistence, with no adjustment for possible cross-sectional effects. However, since our motivation is largely in terms of autoregressive effects, it is useful to provide evidence on the relative roles of the autoregressive vs. the cross-sectional effects on the estimated persistence. We accomplish this by re-running the main regression in Table 2, Panel B in a firm fixed-effects specification, where the resulting coefficients are entirely due to autoregressive persistence effects. The tenor of the results remains largely unchanged with this specification. Specifically, the across-quintiles range in persistence in Table 2, Panel B is 0.427, while the corresponding range is 0.454 in the fixed-effects specification. Thus, the predictive power of earnings volatility for earnings predictability is largely due to autoregressive effects rather than to cross-sectional variation in mean profitability.

Recall that our sample is limited to large firms with December year-end; here we provide evidence on the effect of these restrictions on our results. Fiscal year-end does not seem to matter, the results for non-12/31 firms are much the same as those presented above. Small firms, though, have less reliable relation between volatility and persistence - for firms with assets

[10] We thank an anonymous referee for pointing out this distinction and suggesting the test.

between \$20 million and \$100 million, the range in persistence across volatility quintiles is only about a third to a half that for large firms, and the relation disappears for firms with assets of less than \$20 million. These results, however, should be treated with caution. Our impression from working with the small firms is that this subsample is problematic, e.g., it has a massive outlier problem. In any case, the small firms excluded from our main tests are economically negligible; they comprise less than three percent of the aggregate market value of all firms.

We also provide evidence on the economic and accounting determinants of earnings volatility and its relation to earnings persistence. These results serve two purposes; first, they provide evidence on the construct validity and possible alternative interpretations of the earnings volatility variable; second, they can potentially provide an empirical instrument which avoids the substantial time-series data requirements to compute the earnings volatility variable. On the economic determinants side, we first explore the relation between earnings volatility and industry membership using the Fama-French classification of 12 industry clusters. We find reliable links between industry membership and earnings volatility quintiles, results presented in Table 6, Panel A. Specifically, we find that utility firms strongly cluster and dominate in the lowest volatility quintile, with more than half of the utility firms in the lowest volatility quintile, and more than half of the firms in that quintile being utility firms. There are also reliable clusters in the upper volatility quintiles, with Energy, Health Care, and especially Business Equipment exhibiting a strong presence.

We also explore the relation between earnings volatility and the following variables:

Volatility of Cash flow from operations and Volatility of Sales: Proxies for real economic volatility, predict positive relation with earnings volatility.

Assets, Market value, and Sales: Proxies for size, because of diversification effects predict negative relation with earnings volatility.

Operating cycle: Since longer operating cycles indicate more vulnerability to economic shocks, expect positive relation with earnings volatility.

Mergers and acquisitions: Sign unclear; possible diversification or size effects argue for a negative relation but weakness in targets and integration problems point to a possible positive relation with earnings volatility.

Correlation of Revenues and Expenses: Recall that Dichev and Tang (2008) argue that volatility is increasing with worse matching of revenues and expenses; based on this argument, expect a negative relation with earnings volatility.

R&D levels: Proxy for poor matching and/or involvement in new-economy winner-take-all activities. Predict a positive relation with earnings volatility.

Level of absolute accruals and level of accrual estimation errors (as in Dechow and Dichev 2002): Proxies for accrual quality. Low-quality accruals are expected to manifest as noise in the determination of earnings, leading to higher volatility of earnings.

The mean of these variables across earnings volatility quintiles are presented in Table 6, Panels B and C (results for medians have the same tenor). An inspection of these results reveals that they are largely consistent with expectations, with all variables exhibiting strong economic and statistical associations in predicted direction; the only exception is mergers and acquisitions, where we find little in terms of a reliable economic relation. One upshot from these results is support for the economic and accounting conjectures underpinning the earnings volatility variable. Another upshot is that perhaps these relations can be used to build an instrument for earnings volatility, which avoids the taxing time-series data requirement. Our initial efforts in this direction were not successful. We tested a number of specifications, where earnings volatility is regressed on various combinations of variables, and then the resulting loadings are used to produce the instrument. The explanatory power of these regressions was only moderate, and the resulting proxy was inferior to earnings volatility in capturing earnings persistence.[11]

However, we find that quarterly earnings volatility is an excellent proxy for 5-year annual earnings volatility in terms of its relation to persistence, even without adjusting for seasonal effects. Table 6, Panel D presents the results for one-year ahead predictive regressions of *annual* earnings, based on the volatility of earnings from the most recent four and eight *quarterly* earnings. An examination of Panel D shows that the results using past eight quarters are nearly as good as the benchmark results using past 5 years of annual earnings in Table 2, Panel B. The results using four quarters are just a notch weaker but the overall impression is that using quarterly volatility provides excellent stratification on future earnings persistence. Since the correlation between annual and quarterly earnings volatility is large but far from unity (the Spearman correlation between the 5-year annual volatility and 8 and 4-quarter quarterly volatility is 0.67 and 0.49 respectively), it is possible that quarterly volatility predicts earnings persistence above and beyond annual volatility and there may be gains from combining the predictive power of these two volatility specifications; we leave the investigation of this conjecture for future research.

Finally, we run out-of-sample forecasting tests to corroborate the in-sample estimation results, and to provide additional evidence on the relative utility of the earnings volatility specification versus the other models considered in this study. We rely on mean and median absolute forecast errors as a gauge of forecasting accuracy, where forecast error is equal to the actual earnings realization minus the forecast based on the investigated model (e.g., earnings

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[11] We also explored four other variables as economic determinants of earnings volatility, namely leverage, market-to-book ratio, firm age and price volatility. As expected, firms with high volatility of earnings are younger, have higher market-to-book ratio and higher price volatility. They also have lower leverage, which suggests that volatile firms choose or are restricted to low levels of debt. However, the strength of these relations is generally modest and their consideration does not change any of the other impressions in the paper.

volatility, level of accruals). Forecasts are produced by a rolling-forward estimation, where existing values of the predictive variable are used to split the sample into quintiles and the autoregressive regressions of current on past earnings are used to produce differential persistence coefficients across quintiles, which are then applied to current earnings to produce a forecast of one to five-year ahead earnings. The out-of-sample results yield two impressions. First, consistent with the in-sample evidence, the earnings volatility model produces mean and median forecast errors which are significantly lower than those for the other examined variables, including level of accruals, earnings level, and cash flow volatility. Second, the superiority of the earnings volatility model is concentrated in firms with low to medium volatility of earnings. This result seems useful for practice because these firms dominate the sample in terms of assets and market capitalization (actual results not included but available upon request).

5. Analyst forecasts tests

In this section, we investigate whether financial statement users are aware of the existence and magnitude of the relation between earnings volatility and earnings predictability. We use financial analysts as a proxy for sophisticated users of financial information, and examine whether their forecasts incorporate existing earnings volatility information. We have diffuse priors about the extent to which analysts impound such information. On the one hand, there is substantial evidence that analysts are sophisticated information intermediaries, and thus they are likely to understand the link between earnings volatility and earnings predictability, e.g., see review in Brown (1993). On the other hand, a number of studies identify systematic biases in analysts' forecast errors, which suggests that analysts do not fully impound the implications of existing information, e.g., Frankel and Lee (1998). If analysts do not fully understand and incorporate the relation between earnings volatility and earnings persistence in forecasting future earnings, using earnings volatility information would allow one to identify predictable patterns in their forecast errors. It is also possible that analysts understand the implications of earnings volatility for future earnings but their forecasts still reveal predictable errors because of career or incentive concerns. For example, firms with high volatility earnings are likely to have more information uncertainties and more information asymmetries, so analysts may bias their forecasts, hoping to get preferred access to internal data. For our purposes, we focus on establishing the existence of predictable forecast errors rather than on distinguishing between their possible causes.

We test whether earnings volatility information allows the identification of predictable forecast errors using two complementary specifications, "levels" and "changes", depending on whether we control for level of current earnings or level of current forecast error. Our first specification is conditional on the level of current earnings and uses the same sample and portfolio specification first presented in Figure 3, except for an additional requirement for analyst forecast data. Recall that the advantage of this approach is that the two portfolios are matched on current profitability, while exhibiting a sharp and economically large divergence in future profitability. If analysts impound the implications of earnings volatility for future earnings, their forecasts for the two portfolios in Figure 3 would match the expected divergence in profitability. Table 7 provides the results for this specification. Analyst forecasts are from IBES and are defined as the median earnings forecast made in the first month after the announcement of realized earnings for the current period. Note that Table 7 presents results not only for portfolio means but also for the 10^{th}, 25^{th}, 50^{th}, 75^{th}, and 90^{th} percentile of the empirical distributions of analysts' forecasts and realized earnings, providing an exhaustive account of the properties of the underlying variables. This approach provides a clear illustration of the economic magnitude of the results, and also ensures robustness, which is important for analyst forecasts and related variables exhibiting pronounced non-normalities (Abarbanell and Lehavy 2003).

The first three lines in Table 7 compare the properties of actual current earnings at time t across the two portfolios. The results confirm that the two-pass sorting controls for current earnings nearly perfectly; note that the two portfolios are matched on current earnings not only at the mean and the median but also at all other percentiles of the empirical distributions. Thus, any deviations in future profitability can be fully ascribed to differences in their earnings volatility information. The next three lines in Panel A reflect the properties of the analysts' forecasts for these two portfolios at the one-year-ahead horizon, t+1. An examination of the mean and the percentiles of the empirical distribution reveal that analyst forecasts correctly anticipate that firms with high volatility will have a sharper decline in future earnings than firms with low volatility. However, the predicted divergence in profitability across the low and high volatility portfolios is rather modest, 0.6 percent at the mean, virtually 0 at the median, and a maximum of 1.6 percent at the 10th percentile. In contrast, the divergence in actual realized t+1 earnings (presented in the next three lines) is much more substantial. The difference in realized t+1 earnings is 2 percent at the mean, 0.8 percent at the median, and reaches a high of 4.5 percent at the 10th percentile. The last line in Panel A includes statistical tests for the difference between the forecast and realized earnings differences at the mean and median values of the empirical distributions. The p-value for the difference in mean-differences is 0.001 (from a t-test) and it is 0.015 for the difference in median-differences (from a Wilcoxon test), indicating reliable statistical significance. Thus, analysts' forecasts incorporate only partially the available earnings volatility information. Based on the quantitative magnitude of the differences, the results suggest that analysts incorporate less than half of the full implications of earnings volatility for earnings predictability. Additional tests reveal that the results are nearly the same for two-year-ahead earnings forecasts (actual results not included).

Our second analyst forecast specification controls for the magnitude of the current forecast error. If analysts fail to recognize that earnings are less persistent for high volatility firms, high volatility firms with positive (negative) earnings surprises at time t are expected to have *negative (positive)* earnings surprise at t + 1. However, existing research shows that analyst forecast errors at time t are positively correlated with analyst forecast errors at time t+1, implying that firms with positive (negative) current earnings surprises are expected to have positive (negative) earnings surprise in the future (e.g., Abarbanell and Bernard 1992 and Ali, Klein, and Rosenfeld 1992). Thus, in the empirical analysis, it is essential to control for the magnitude and sign of earnings surprises at time t in the prediction of forecast errors at t+1. More specifically, we use the following regression to test whether analysts fully incorporate the relation between earnings volatility and earnings persistence:

$$FE_{t+1} = b_1 + b_2*High\_vol_t + b_3*FE_t + b_4*High\_vol_t*FE_t + \varepsilon_t$$

In this model, *FE* denotes forecast error and $High\_vol_t$ is a dummy variable, which is coded as 1 if a firm is in the top quintile of earnings volatility at time t and 0 if the firm is in the bottom quintile of earnings volatility. To maintain continuity with the preceding specifications in the paper and to maximize the power of the test, we only include firms in the highest and lowest quintiles of earnings volatility in the regression.

Analysts forecast errors for year t are defined as the actual year t IBES earnings minus the last median analyst forecast for year t prior to the announcement of year t earnings. Analyst forecast errors for t+1 are defined as actual IBES earnings for t +1 minus the first median analyst forecast for t +1 made immediately after the announcement of year t earnings. The variable of interest is the interaction term between current forecast errors and the indicator variable for firms in the highest earnings volatility quintile. The coefficient on the interaction variable is expected to be negative if analysts fail to fully incorporate the information in earnings volatility for future earnings. Panel A of Table 8 presents the descriptive statistics for forecast errors at t and t+1. The mean (median) forecast error is nearly zero for time t and -0.65% (-0.20%) for t+1, indicating unbiased short-term forecasts and somewhat optimistic longer-term forecasts, in line with existing results. In addition, the mean (median) t+1 forecast error for firms in the lowest earnings volatility quintile is close to zero, while it is -1.16% (-0.68%) for firms in the highest volatility quintile. This evidence indicates that analysts are much more optimistic for high volatility firms, which needs to be kept in mind in interpreting the results.

Panel B of Table 8 presents the regression results. As expected from the descriptive statistics above, the intercept term for low volatility firms is nearly zero, and it is reliably negative for high volatility firms. The slope coefficient on the forecast error variable for low volatility firms is 0.741 (p-value = 0.001), indicating high persistence of forecast errors for these firms. The coefficient on the high volatility indicator variable is -0.571 (p-value = 0.013), which suggests that the forecast error persistence for high volatility firms is much lower than that for the low volatility firm benchmark. A comparison of the economic magnitude of these coefficients indicates that the consideration of earnings volatility information provides key insights into the properties of analysts forecast errors. At a magnitude of 0.170 (0.741 – 0.571), the error persistence of high volatility firms is less than a quarter of the error persistence of low volatility firms. This evidence confirms that analysts do not fully understand the implications of existing earnings volatility information for future earnings, and thus conditioning on such information allows one to identify reliable and economically important patterns in analyst forecast errors.[12]

Figure 4 provides a graphical view of the regression results, plotting t+1 forecast errors (Y axis) as a function of time t forecast errors (X axis), conditional on earnings volatility. Specifically, the full analyst forecast sample is first sorted into 5 quintiles on the level of earnings volatility, and then firms within the bottom and top earnings volatility quintiles are further sorted into 50 portfolios on the level of the current analyst forecast error. Figure 4 plots the medians and the slope lines for these portfolios, using triangles and solid line for low volatility observations and dots and dashed line for the high volatility observations. An examination of Figure 4 provides compelling visual evidence that the persistence of analyst forecast errors differs dramatically conditional on earnings volatility. The triangle observations for low volatility firms cluster tightly along a steep solid line as compared to a much more diffuse cloud of dots and much flatter dashed line for high volatility firms.

Summarizing, two different test specifications reveal that conditioning on current earnings volatility information allows for the identification of predictable and economically large analyst forecast errors. These results suggest that analysts do not fully understand the implication of earnings volatility for earnings predictability.

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[12] To put the different slope coefficient for low volatility and high volatility firms into perspective, we also run an unconditional regression of future on current forecast errors for the full sample of 7,290 observations, and obtain a slope coefficient of 0.22. This magnitude for the slope coefficient is similar to those obtained in prior studies (e.g., the persistence coefficient is 0.26 in Ali, Klein and Rosenfeld 1992 and 0.20 in Abarbanell and Bernard 1992). A comparison of these unconditional results with the conditional results in Panel B of Table 8 indicates that firms with low volatility earnings have much higher slope coefficient than the average firm, while high volatility firms have only slightly lower slope coefficient than the average firm. These results essentially suggest that existing evidence understates the persistence of analyst forecast errors because the comparatively low estimates of persistence in such studies are due to the low persistence for high volatility firms, and the fact that high volatility firms by their nature dominate the sample variation of magnitude of forecast error. The implication of our results is that by stratifying on earnings volatility one can create samples where the persistence of forecast errors is much higher than previously thought.

6. Conclusion

This paper investigates the link between earnings volatility and earnings predictability. The motivation for this investigation relies on recent survey evidence which reveals widespread managerial beliefs that higher volatility indicates lower earnings predictability. In addition, a consideration of existing results offers some clues about the potential mechanism behind this relation including economic and accounting determinants. The empirical results reveal that earnings volatility provides reliable discrimination on relative earnings persistence and predictability up to five-years ahead, and dominates in strength existing results like the accrual effect and the extreme-earnings mean-revert effect. We also find that analysts make systematic errors in their interpretations of earnings volatility information, incorporating less than half of its full implications. These findings open a number of possibilities for future research. One potential direction is to expand and solidify these results using other samples and variable definitions. Another future direction is exploring the link between the identified fundamental relations and derived or observed equity values.

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Appendix

A1. The effect of transitory components on earnings persistence

Consider a company with earnings consisting of two components $E_t = X_t + Z_t$ where $X_t = \gamma X_{t-1} + \varepsilon_t$ and $Z_t = \delta Z_{t-1} + \eta_t$, such that $\varepsilon_t \perp \eta_t$. Further assume that X_t is more persistent than Z_t, i.e., $\gamma > \delta$

Running the regression

$$E_t = b_0 + b_1 E_{t-1} + \mu_t \tag{1}$$

will result in a slope coefficient with the following expected value:

$$\begin{aligned} E(b_1) &= \frac{Cov(X_t + Z_t, X_{t-1} + Z_{t-1})}{Var(X_{t-1} + Z_{t-1})} = \frac{Cov(\gamma X_{t-1} + \varepsilon_t + \delta Z_{t-1} + \eta_t, X_{t-1} + Z_{t-1})}{Var(X_{t-1} + Z_{t-1})} \\ &= \gamma \frac{Var(X_{t-1})}{Var(X_{t-1} + Z_{t-1})} + \delta \frac{Var(Z_{t-1})}{Var(X_{t-1} + Z_{t-1})} = \gamma(1-\omega) + \delta\omega \end{aligned} \tag{2}$$

where w is the fraction of variance attributable to the less persistent components of earnings. In high volatility environments w will be larger, so it immediately follows that the estimated slope b_1 (and adjusted R-squared) will be lower in such circumstances (because $\gamma > \delta$).

A2. Effects of time-series autoregressive and cross-sectional effects on the estimated persistence coefficient

Suppose earnings E_{it} consist of a firm level μ_i (e.g., time series mean) which has cross-sectional variance Var(μ_i) = σ_u^2 and the autoregressive process X_{it} with firm-specific variance Var(X_{it}) = σ_i^2. Specifically

$$X_{it} = bX_{it-1} + \eta_{it} \text{ and } E_{it} = \mu_i + X_{it} \text{ (for simplicity} \mu_i \perp X_{it}) \tag{1}$$

Running the regression using panel data

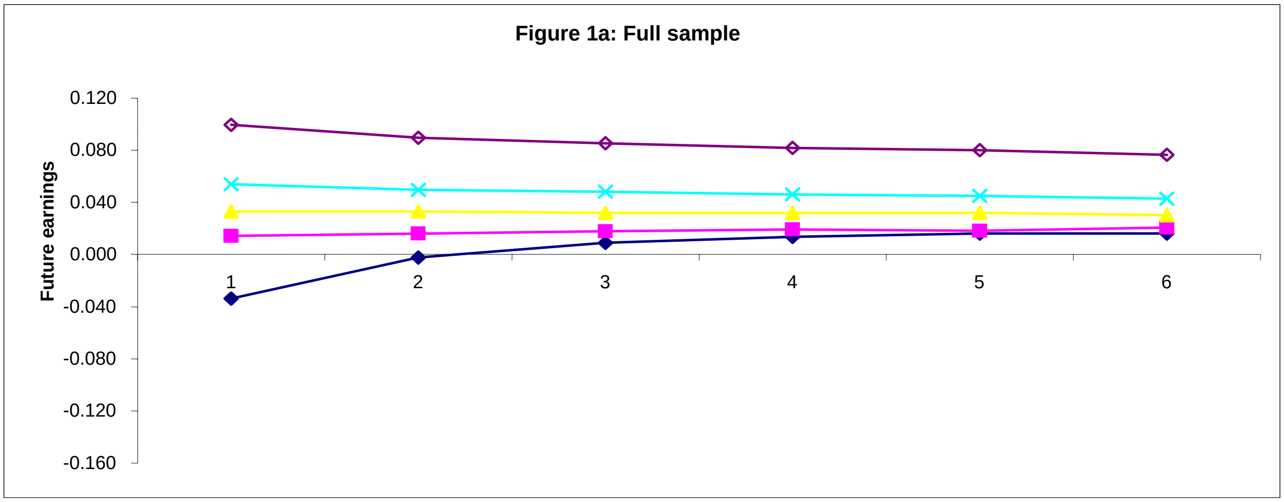
$$E_t = b_0 + b_1 E_{t-1} + \varepsilon \tag{2}$$

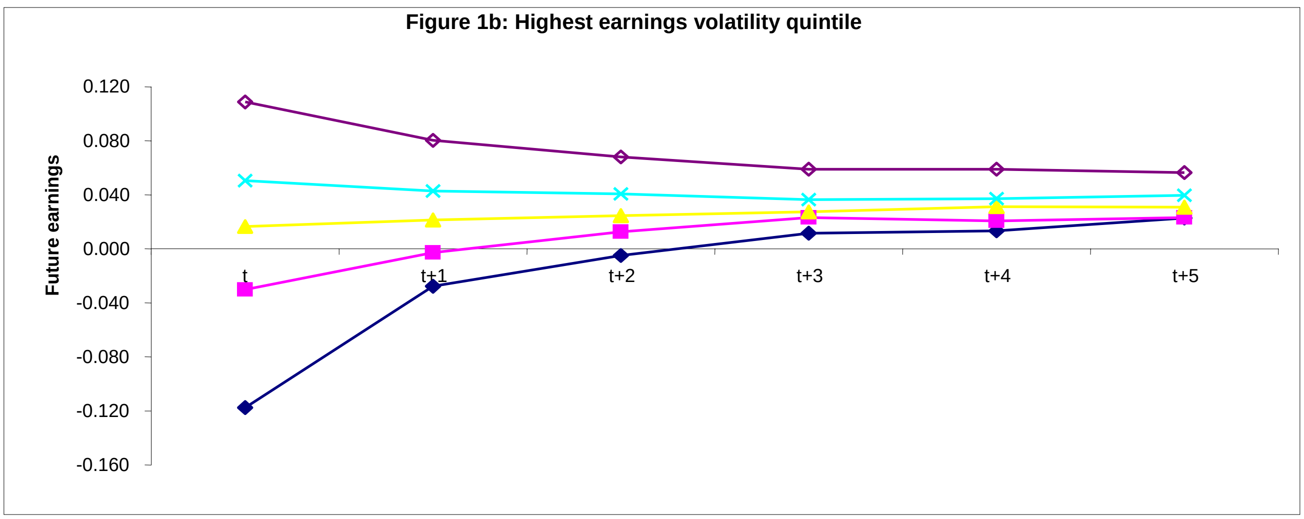
Will result in the following expectation for the slope coefficient

$$E(b_1) = \frac{Cov(\mu_i + X_{it}, \mu_i + X_{it-1})}{Var(\mu_i + X_{it-1})} = \frac{\sigma_\mu^2 + \beta\sigma_i^2}{\sigma_\mu^2 + \sigma_i^2} \tag{3}$$

Equation (3) indicates that the resulting slope coefficient is a weighted-average of the time-series autoregressive persistence (within firms) and cross-sectional (between firms) variation in mean profitability.

Figure 1
Mean reversion of five-year future earnings conditional on earnings volatility





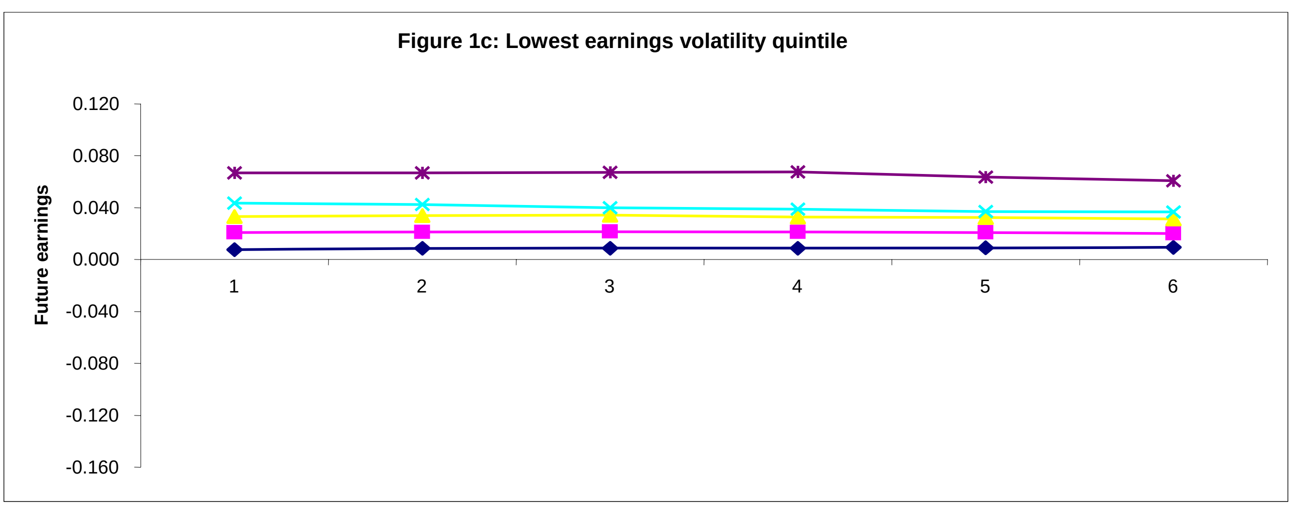
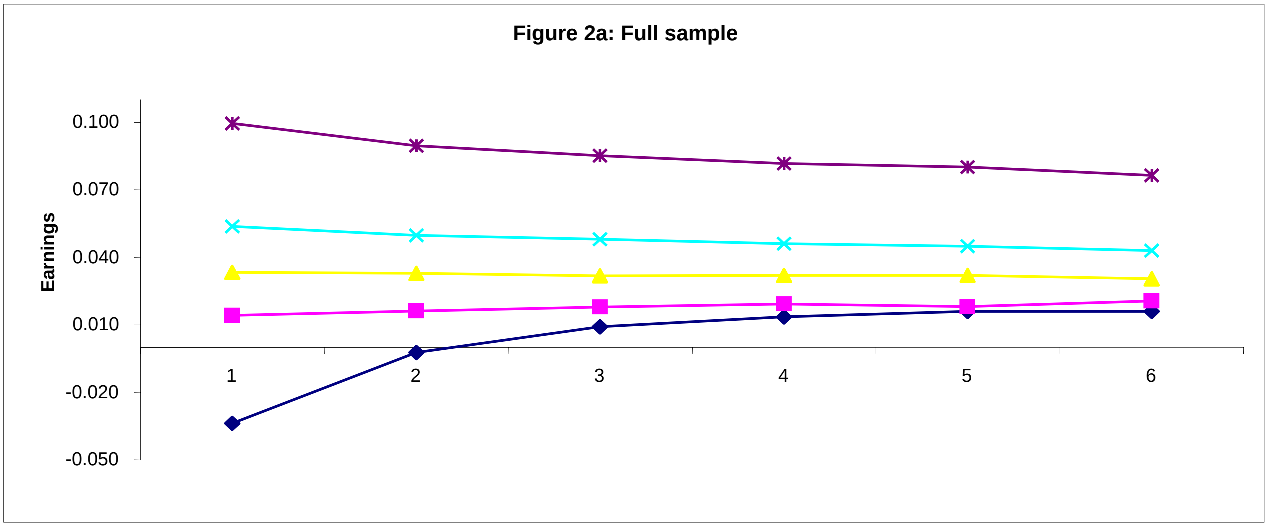
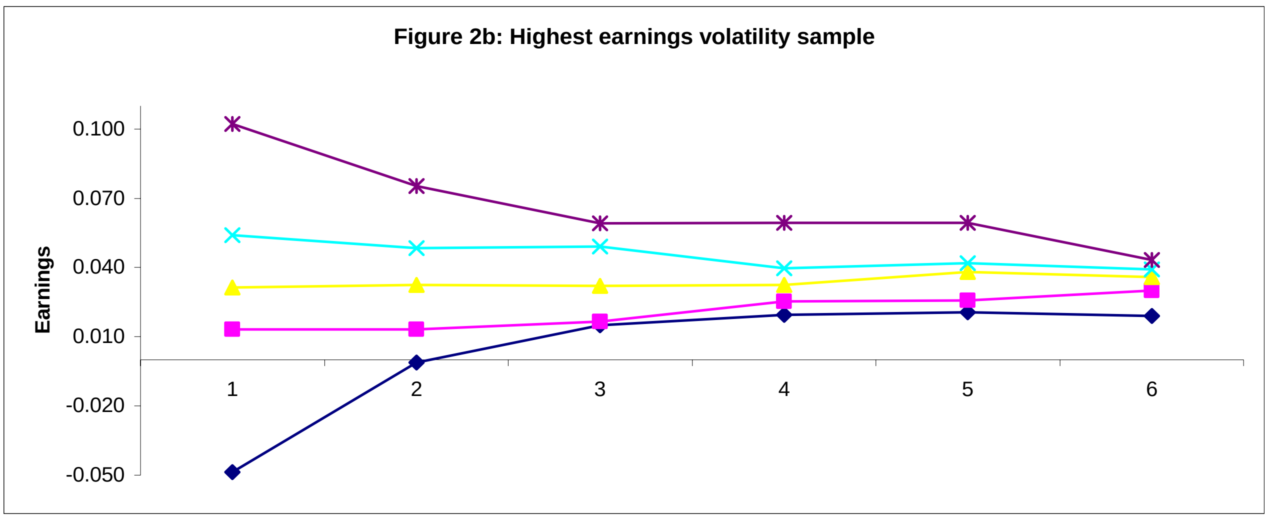


Figure 1 (continued)

In Figure 1a, the full sample is sorted into five quintiles by the level of current earnings. The graph for the full sample plots the median current earnings and future earnings for each quintile. In Figure 1b and Figure 1c, the full sample is first sorted into five quintiles by the level of earnings volatility. Then the observations within the highest (lowest) earnings volatility quintile are sorted into five quintiles by the level of current earnings. The graph for highest (lowest) earnings volatility plots the median current earnings and future earnings for each quintile. *Current Earnings* is defined as the earnings before extraordinary item (Compustat Annual Item 123) deflated by the average total assets (Compustat Item 6). *Future Earnings* is future earnings over the next five years.

Figure 2
Mean reversion of five-year future earnings conditional on earnings volatility and controlling for the dispersion of current earnings





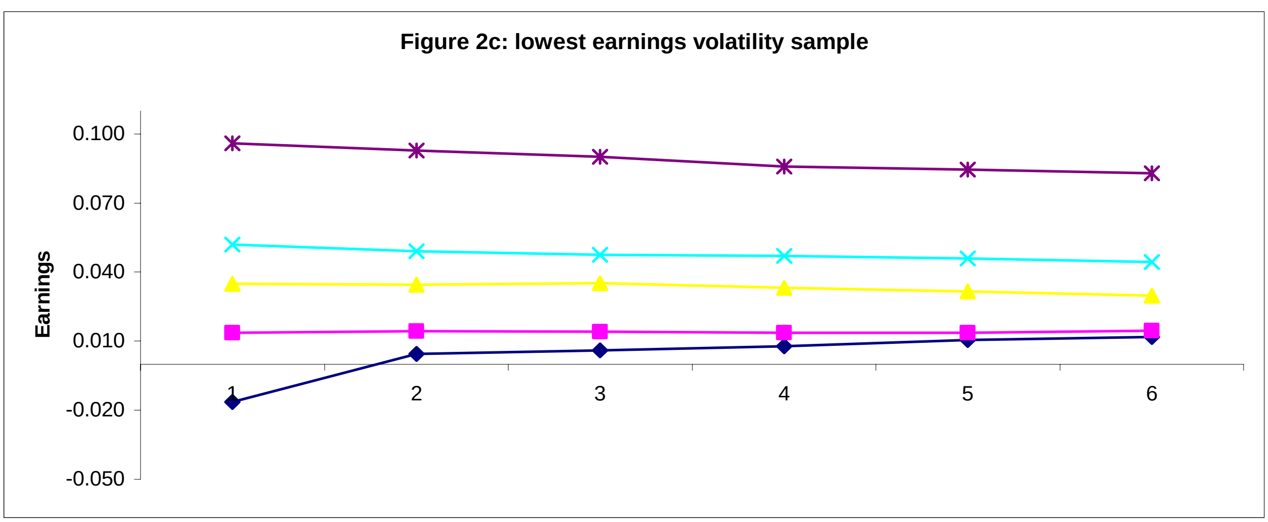
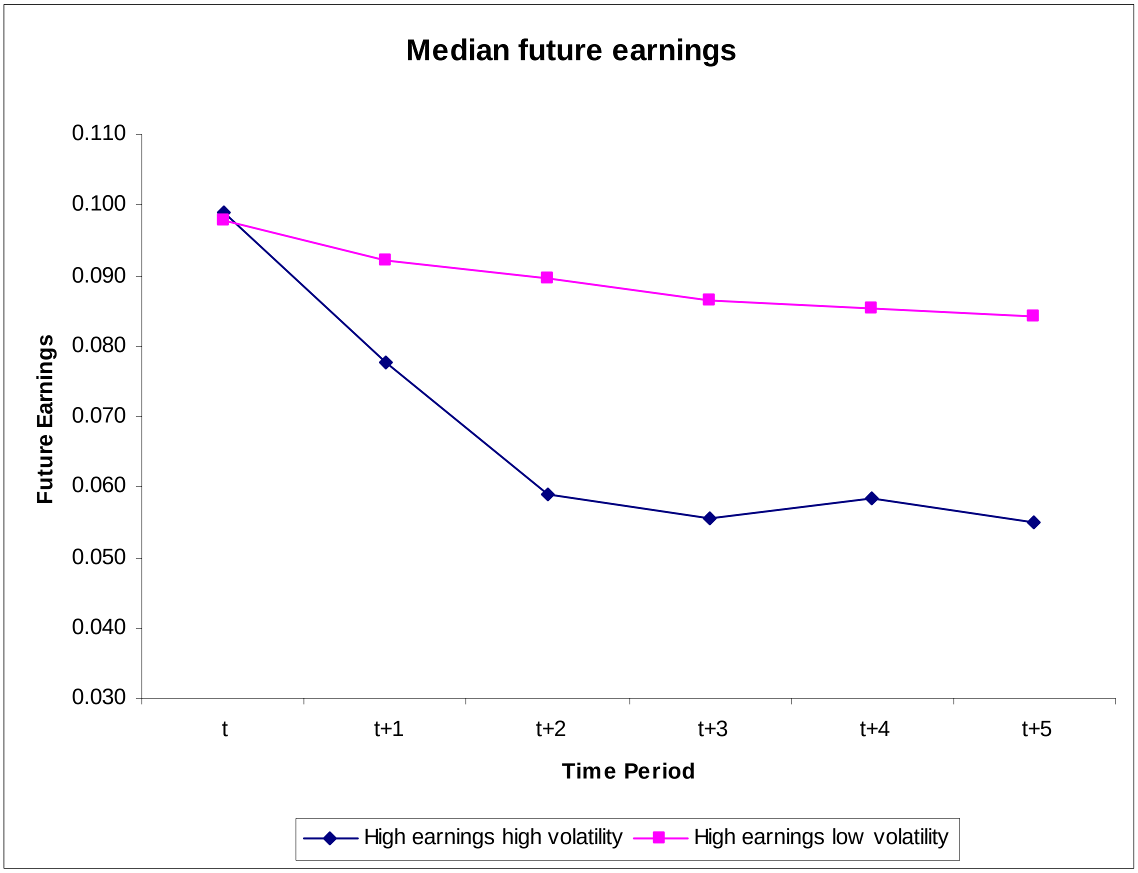


Figure 2 (continued)

In Figure 2a, the full sample is sorted into five quintiles by the level of current earnings. The graph for the full sample plots the median current earnings and future earnings for each quintile. The following steps are involved to produce the graph for the highest (lowest) earnings volatility sample in Figure 2b and Figure 2c,. First, the full sample is sorted into 20 portfolios by the level of current earnings. Five earnings volatility quintiles are formed within each 20 current earnings portfolios. Combining each of the highest (lowest) earnings volatility quintiles from the 20 current earnings portfolios together forms the highest (lowest) earnings volatility sample. Then the observations in the highest (lowest) earnings volatility sample are sorted into five quintiles by the level of current earnings. The graph for the highest (lowest) earnings volatility plots the median current earnings and future earnings for each quintile. *Current Earnings* is defined as the earnings before extraordinary item (Compustat Annual Item 123) deflated by the average total assets (Compustat Item 6). *Future Earnings* is future earnings over the next five years.

Figure 3
Five-year future earnings for portfolios constructed to control for current earnings



The two portfolios are constructed in the following way. First, the full sample is sorted into 20 portfolios on the level of current earnings. Within each earnings portfolio, quintiles of earnings volatility are formed. The high earnings high volatility sub-sample includes observations from the intersection of the highest earnings volatility quintile and earnings level portfolios 17 to 20. The high earnings low volatility sub-sample includes observations from the intersection of the lowest earnings volatility quintile and earnings level portfolios 17 to 20. Portfolio 1 includes the high earnings high volatility sub-sample. Portfolio 2 includes the high earnings low volatility sub-sample. *Current Earnings* is defined as the earnings before extraordinary item (Compustat Annual Item 123) deflated by the average total assets (Compustat Item 6). *Future Earnings* is future earnings over the next five years.

Figure 4
Persistence of analyst forecast errors conditional on earnings volatility

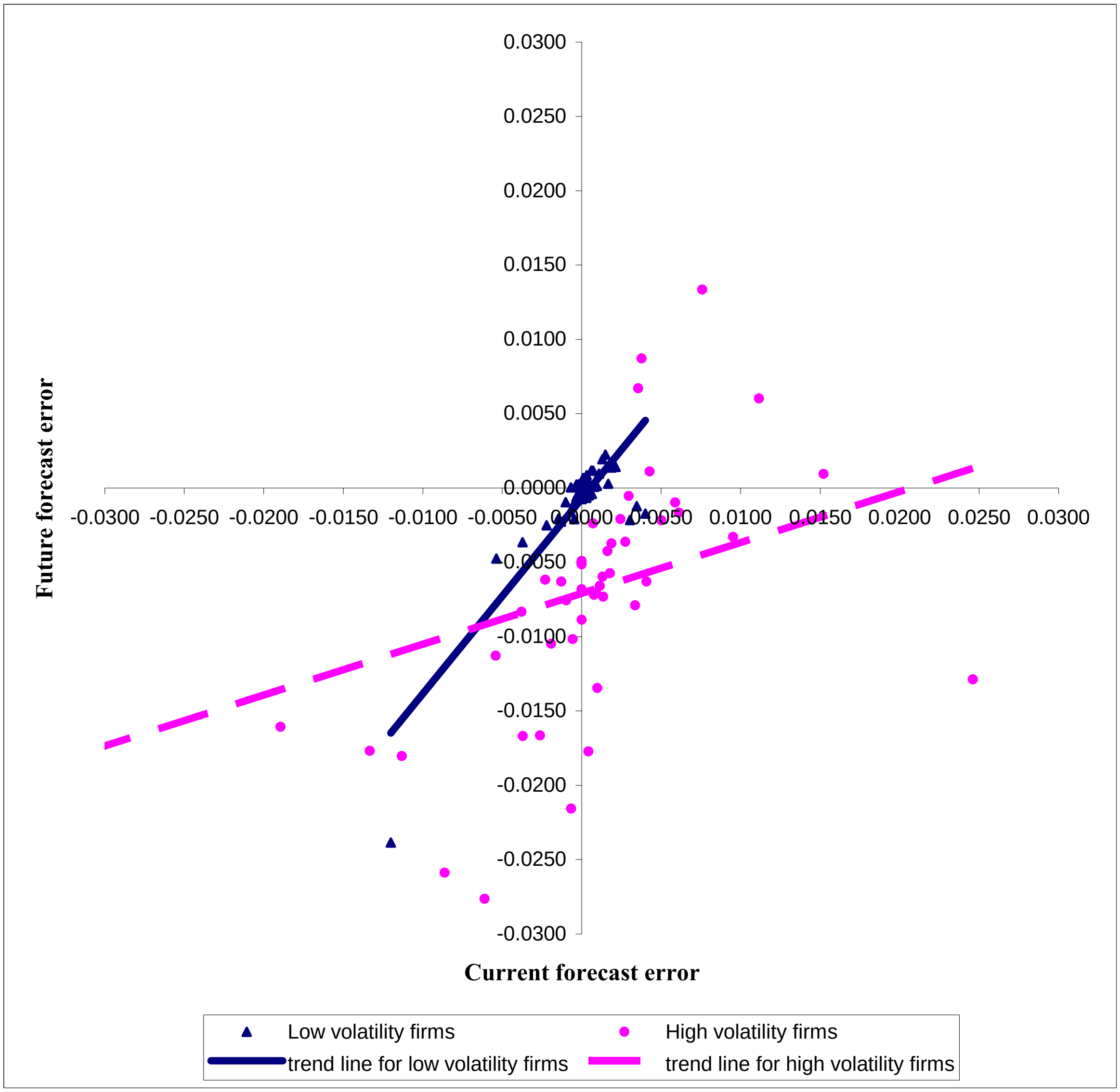


Figure 4 is constructed in the following way. First, the full sample of 7,290 analyst forecasts is sorted into 5 quintiles on the level of earnings volatility. Second, firms in the lowest earnings volatility quintile are sorted into 50 portfolios on the level of the current analyst forecast error. The median current forecast error (time t) and the median future forecast error (time t+1) for each portfolio are plotted in the figure as triangles, with a solid line indicating the slope for these observations. The same procedure is repeated for firms in the highest earnings volatility quintile, with 50 portfolios depicted as dots determining a dashed slope line.

Table 1
Derivation of the sample and descriptive statistics

Panel A: Derivation of the sample

Compustat firm-years over 1984 to 2004 with 12/31 fiscal year-end, and available total assets, cash flows from operations and earnings	121,482
Firm-years with available deflated earnings, cash flows and accruals	79,879
Firm-years with assets >= $100 millions	44,519
Firm-years with available data on earnings volatility and cash flow volatility (based on the most recent five years)	22,990
Firm-years remaining after truncating the top and bottom 1 percent on all variables	22,113
Firm-years in the final sample	**22,113**

Panel B: Descriptive statistics

Variables	N	Mean	Std Dev	Minimum	Maximum
Earnings	22,113	0.031	0.066	-0.513	0.294
Accruals	22,113	-0.055	0.062	-0.516	0.203
|Accruals|	22,113	0.063	0.053	0.000	0.516
CFOs	22,113	0.085	0.069	-0.230	0.378
Vol(Earnings)	22,113	0.040	0.157	0.000	12.448
Vol(CFOs)	22,113	0.039	0.035	0.000	0.632

Earnings is defined as earnings before extraordinary item (Compustat Annual Item 123) deflated by average total assets (Compustat Item 6). *CFOs* is defined as the cash flow from operating activities (Compustat Item 308) deflated by average total assets. *Accruals* is calculated as the difference between Earnings and CFOs. |*Accruals*| is the absolute amount of Accruals. *Vol(Earnings)* is defined as the firm-specific volatility of earnings, which is calculated as the standard deviation of *Earnings* over the most recent 5 years. *Vol(CFOs)* is defined as the firm-specific volatility of cash flows from operations, which is calculated as the standard deviation of *CFOs* over the most recent 5 years.

Table 2
Results for the earnings persistence regression:
$Earnings_{t+1} = \alpha + \beta * Earnings_t + \varepsilon$

Panel A: Regression result for the full sample

	β (Persistence)	Adj. R^2
Full sample	0.652	0.398

Panel B: Regression results by quintiles of earnings volatility

Quintiles by *Vol(Earnings)*	β (Persistence)	Adj. R^2
Quintile 1	0.934	0.704
Quintile 2	0.888	0.570
Quintile 3	0.838	0.463
Quintile 4	0.755	0.414
Quintile 5	0.507	0.296
Difference (Quintile 1 - Quintile 5)	0.427	0.408
p-value on Difference	*< 0.001*	*<0.001*

Panel C: Regression results by quintiles of absolute amount of accruals

Quintiles by *|Accruals|*	β (Persistence)	Adj. R^2
Quintile 1	0.870	0.502
Quintile 2	0.804	0.421
Quintile 3	0.818	0.430
Quintile 4	0.783	0.423
Quintile 5	0.545	0.385
Difference (Quintile 1 - Quintile 5)	0.325	0.116
p-value on Difference	*< 0.001*	*<0.001*

Panel D: Regression results by quintiles of earnings level

Quintiles by *Earnings*	**β (Persistence)**	**Adj. R^2**
Quintile 1	0.793	0.031
Quintile 2	0.877	0.133
Quintile 3	0.853	0.242
Quintile 4	0.790	0.379
Quintile 5	0.620	0.540
Difference (Quintile 1 - Quintile 5)	0.179	- 0.509
p-value on Difference	*< 0.001*	*<0.001*

Panel E: Regression results by quintiles of cash flow volatility

Quintiles by *Vol(CFOs)*	**β (Persistence)**	**Adj. R^2**
Quintile 1	0.785	0.527
Quintile 2	0.755	0.494
Quintile 3	0.663	0.404
Quintile 4	0.618	0.389
Quintile 5	0.609	0.342
Difference (Quintile 1 - Quintile 5)	0.176	0.185
p-value on Difference	*< 0.001*	*<0.001*

All β (Persistence) coefficients are significant at the 0.001 level. The p-value for the difference in persistence coefficients across quintiles is derived from a t-test. The p-value for the difference in the Adj. R^2 across quintiles is derived from a bootstrap test (see text for full details). $Earnings_t$ is defined as earnings before extraordinary item (Compustat Annual Item 123) deflated by the average total assets (Compustat Item 6). |*Accruals*| is the absolute amount of Accruals. *Vol(Earnings)* is defined as the firm-specific standard deviation of *Earnings* over the most recent 5 years. *CFOs* is defined as the cash flow from operating activities (Compustat Item 308) deflated by average total assets. *Vol(CFOs)* is defined as the firm-specific standard deviation of *CFOs* over the most recent 5 years.

Table 3
The implications of earnings volatility for long-term earnings

Panel A: Regression results for the full sample

	β	**Adj. R^2**	**Number of observations**
$\text{Earnings}_{t+1} = \alpha + \beta^*\text{Earnings}_t$	0.652	0.398	9,102
$\text{Earnings}_{t+2} = \alpha + \beta^*\text{Earnings}_t$	0.484	0.211	7,502
$\text{Earnings}_{t+3} = \alpha + \beta^*\text{Earnings}_t$	0.449	0.168	6,147
$\text{Earnings}_{t+4} = \alpha + \beta^*\text{Earnings}_t$	0.421	0.139	5,011
$\text{Earnings}_{t+5} = \alpha + \beta^*\text{Earnings}_t$	0.381	0.106	4,032

Panel B: Regression results for the highest earnings volatility quintile

	β	**Adj. R^2**	**Number of observations**
$\text{Earnings}_{t+1} = \alpha + \beta^*\text{Earnings}_t$	0.507	0.296	1,682
$\text{Earnings}_{t+2} = \alpha + \beta^*\text{Earnings}_t$	0.287	0.104	1,314
$\text{Earnings}_{t+3} = \alpha + \beta^*\text{Earnings}_t$	0.261	0.088	1,029
$\text{Earnings}_{t+4} = \alpha + \beta^*\text{Earnings}_t$	0.241	0.069	822
$\text{Earnings}_{t+5} = \alpha + \beta^*\text{Earnings}_t$	0.177	0.031	659

Panel C: Regression results for the lowest earnings volatility quintile

	β	**Adj. R^2**	**Number of observations**
$\text{Earnings}_{t+1} = \alpha + \beta^*\text{Earnings}_t$	0.934	0.704	1,899
$\text{Earnings}_{t+2} = \alpha + \beta^*\text{Earnings}_t$	0.867	0.561	1,620
$\text{Earnings}_{t+3} = \alpha + \beta^*\text{Earnings}_t$	0.774	0.359	1,365
$\text{Earnings}_{t+4} = \alpha + \beta^*\text{Earnings}_t$	0.812	0.387	1,140
$\text{Earnings}_{t+5} = \alpha + \beta^*\text{Earnings}_t$	0.805	0.315	922

Table 3 (continued)

Panel D: Regression results for the highest earnings volatility quintile, controlling for the level of current earnings

	β	**Adj. R^2**	**Number of observations**
$Earnings_{t+1} = \alpha + \beta * Earnings_t$	0.493	0.215	1,684
$Earnings_{t+2} = \alpha + \beta * Earnings_t$	0.242	0.051	1,292
$Earnings_{t+3} = \alpha + \beta * Earnings_t$	0.303	0.062	963
$Earnings_{t+4} = \alpha + \beta * Earnings_t$	0.254	0.040	716
$Earnings_{t+5} = \alpha + \beta * Earnings_t$	0.126	0.006	538

Panel E: Regression results for the lowest earnings volatility quintile, controlling for the level of current earnings

	β	**Adj. R^2**	**Number of observations**
$Earnings_{t+1} = \alpha + \beta * Earnings_t$	0.812	0.673	1,951
$Earnings_{t+2} = \alpha + \beta * Earnings_t$	0.752	0.532	1,720
$Earnings_{t+3} = \alpha + \beta * Earnings_t$	0.652	0.413	1,512
$Earnings_{t+4} = \alpha + \beta * Earnings_t$	0.607	0.336	1,304
$Earnings_{t+5} = \alpha + \beta * Earnings_t$	0.630	0.343	1,099

All β coefficients are statistically significant at the 0.001 level. *Earnings* is defined as earnings before extraordinary item (Compustat Annual Item 123) deflated by average total assets (Compustat Item 6). *Vol(Earnings)* is the firm-specific standard deviation of *Earnings* over the most recent 5 years. $Earnings_t$ is current year *Earning,* $Earnings_{t+1}$ is the one-year ahead *Earnings,* and so on.

Table 4
The implications of earnings volatility for the sum of earnings over the next five years: Σ (Earnings $_{t+1}$ to Earnings $_{t+5}$) = $\alpha + \beta$*Earnings$_t$ + ε

Panel A: Regression results for the full sample

β	Adj. R^2	Number of observations
2.359	0.311	4,032

Panel B: Regression results for the highest earnings volatility quintile

β	Adj. R^2	Number of observations
1.372	0.172	659

Panel C: Regression results for the lowest earnings volatility quintile

β	Adj. R^2	Number of observations
4.205	0.633	922
2.933 (Difference from Panel B)	*0.461 (Difference from Panel B)*	
<0.001 (P-value of Difference)	*<0.001 (P-value of Difference)*	

All β coefficients are statistically significant at the 0.001 level. *Earnings* is defined as earnings before extraordinary item (Compustat Annual Item 123) deflated by average total assets (Compustat Item 6). *Vol(Earnings)* is the firm-specific standard deviation of *Earnings* over the most recent 5 years. *Earnings$_t$* is current year *Earnings.* *Earnings$_{t+1}$* is the one-year ahead *Earnings,* and so on. The p-value for the difference in persistence and the Adj. R^2 across panels is derived from a bootstrap test (see text for full details).

Table 5
Robustness checks on the influence of past earnings persistence on the documented results

Panel A: Regression results from the model $Earnings_{t+1} = \alpha + \beta*Earnings_t + \varepsilon$ by quintiles of AR(1) residual volatility

Quintiles by *Vol(Residual)*	β (Persistence)	Adj. R^2
Quintile 1	0.839	0.606
Quintile 2	0.889	0.607
Quintile 3	0.823	0.502
Quintile 4	0.727	0.381
Quintile 5	0.521	0.285
Difference (Quintile 1 - Quintile 5)	0.318	0.321
p-value on Difference	*< 0.001*	*<0.001*

Panel B: Earnings persistence using independent sorting on past persistence and past volatility of earnings

Quintiles by Vol(Earnings) / Quintiles by Past Persistence	Quintile 1	Quintile 2	Quintile 3	Quintile 4	Quintile 5	Difference Quintile (1–5)
Quintile 1	0.958	0.859	0.767	0.797	0.432	0.526
Quintile 2	0.907	0.889	0.997	0.725	0.486	0.421
Quintile 3	0.941	0.905	0.720	0.678	0.607	0.334
Quintile 4	0.968	0.886	0.915	0.823	0.507	0.461
Quintile 5	0.904	0.917	0.825	0.724	0.503	0.401
Difference Quintile (1 – 5)	0.054	-0.058	0.058	0.073	-0.071	

All β (Persistence) coefficients are significant at the 0.001 level. The p-value for the difference in persistence coefficients across quintiles is derived from a t-test. The p-value for the difference in the Adj. R^2 across quintiles is derived from a bootstrap test (see text for full details). *Earnings$_t$* is defined as earnings before extraordinary item (Compustat Annual Item 123) deflated by the average total assets (Compustat Item 6). *Vol(Earnings)* is the firm-specific standard deviation of *Earnings* over the most recent 5 years. Past persistence is the persistence coefficient from the model $Earnings_t = \alpha + \beta*Earnings_{t-1}$ using the most recent 5 years. *Vol(Residual)* is defined as the firm-specific standard deviation of the residuals from the above model.

Table 6
Determinants of Earnings Volatility

Panel A: Distribution of 12 Fama-French industries across earnings volatility quintiles

INDUSTRY Row Percentage% *Column Percentage%*	**Quintile of Vol(Earnings) = 1**	**Quintile of Vol(Earnings) = 2**	**Quintile of Vol(Earnings) = 3**	**Quintile of Vol(Earnings) = 4**	**Quintile of Vol(Earnings) = 5**
Consumer Nondurables	7.09 *3.36*	21.7 *7.74*	28.26 *9.04*	23.29 *6.95*	19.66 *5.7*
Consumer Durables	5.59 *1.26*	19.55 *3.32*	28.68 *4.37*	29.42 *4.18*	16.76 *2.31*
Manufacturing	4.79 *5.7*	17.06 *15.3*	25.31 *20.35*	28.66 *21.49*	24.18 *17.62*
Energy	0.93 *0.46*	13.67 *5.12*	25.32 *8.5*	29.7 *9.3*	30.38 *9.24*
Chemicals	4.83 *1.38*	13.91 *3*	23.57 *4.56*	35.72 *6.45*	21.96 *3.85*
Business Equipment	3.73 *2.35*	9.92 *4.71*	14.78 *6.29*	19.91 *7.9*	51.66 *19.93*
Telecommunications	8.36 *4.07*	20.86 *7.65*	22.93 *7.54*	24.4 *7.48*	23.45 *6.99*
Utilities	53.78 *58.47*	27.51 *22.53*	10.57 *7.77*	5.48 *3.75*	2.66 *1.77*
Retail & Wholesale	14.83 *7.84*	23.39 *9.32*	23.55 *8.42*	18.95 *6.32*	19.27 *6.24*
Health Care	6.19 *2.52*	13.61 *4.17*	18.76 *5.16*	26.39 *6.77*	35.05 *8.73*
Finance	12.34 *1.59*	27.92 *2.72*	17.21 *1.5*	18.18 *1.48*	24.35 *1.93*
Others	10.12 *10.99*	17.61 *14.41*	22.47 *16.5*	26.22 *17.94*	23.59 *15.69*

Table 6 (continued)

Panel B: Means of economic factors by quintiles of earnings volatility

Quintiles by *Vol(Earnings)*	Mean Vol(CFO)	Mean Vol(Sales)	Mean M&A Indicator	Mean Assets	Mean Market value	Mean Sales	Mean Operating Cycle
Quintile 1	0.019	0.052	0.062	20750	5153	3529	92.8
Quintile 2	0.029	0.087	0.101	7227	6138	4927	113.2
Quintile 3	0.036	0.110	0.103	6026	6921	4646	120.7
Quintile 4	0.046	0.149	0.101	4954	5082	4123	123.0
Quintile 5	0.066	0.184	0.100	2748	3229	2089	137.2
Difference (Quintile 1 - Quintile 5)	-0.046	-0.133	-0.038	18002	1923	1440	44.4
p-value on Difference	*0.001*	*0.001*	*0.001*	*0.001*	*0.001*	*0.001*	*0.001*

Panel C: Means of accounting factors by quintiles of earnings volatility

Quintiles by *Vol(Earnings)*	Mean R&D/Sales	Mean Corr(Revenue, Expenses)	Mean Abs(accruals)	Mean Std (accrual estimation errors)
Quintile 1	0.004	0.940	0.041	0.009
Quintile 2	0.009	0.904	0.052	0.017
Quintile 3	0.015	0.877	0.059	0.022
Quintile 4	0.070	0.838	0.069	0.030
Quintile 5	0.061	0.647	0.095	0.046
Difference (Quintile 1 - Quintile 5)	-0.057	0.292	-0.054	-0.037
p-value on Difference	*0.001*	*0.001*	*0.001*	*0.001*

Table 6 (continued)

Panel D: Quarterly earnings volatility as a proxy for annual earnings volatility

Annual earnings persistence regressions $Earnings_{t+1} = \alpha + \beta * Earnings_t + \varepsilon$ **by quintiles of quarterly volatility of earnings based on the most recent four quarters**

Quintiles by *Vol(Earn) (4 quarters)*	**β (Persistence)**	**Adj. R^2**
Quintile 1	0.872	0.595
Quintile 2	0.854	0.484
Quintile 3	0.831	0.461
Quintile 4	0.805	0.418
Quintile 5	0.460	0.272
Difference (Quintile 1 - Quintile 5)	0.412	0.323
p-value on Difference	*< 0.001*	*<0.001*

Annual earnings persistence regressions $Earnings_{t+1} = \alpha + \beta * Earnings_t + \varepsilon$ **by quintiles of quarterly volatility of earnings based on the most recent eight quarters**

Quintiles by *Vol(Earn) (8 quarters)*	**β (Persistence)**	**Adj. R^2**
Quintile 1	0.884	0.637
Quintile 2	0.881	0.509
Quintile 3	0.842	0.469
Quintile 4	0.765	0.401
Quintile 5	0.458	0.259
Difference (Quintile 1 - Quintile 5)	0.426	0.378
p-value on Difference	*< 0.001*	*<0.001*

Table 7
Comparison of analysts' forecasts of one-year ahead earnings and realized one-year ahead earnings, conditional on earnings volatility

Realized earnings at time t			Percentiles of the empirical distribution				
Portfolios	N	Mean	10%	25%	50%	75%	90%
High Earnings/Low Volatility	424	0.115	0.077	0.089	0.106	0.137	0.164
High Earnings/High Volatility	425	0.118	0.079	0.091	0.107	0.137	0.171
Difference		*-0.003*	*-0.002*	*-0.002*	*-0.001*	*0.000*	*-0.007*
Analyst forecasts for t+1							
Portfolios	N	Mean	10%	25%	50%	75%	90%
High Earnings/Low Volatility	424	0.129	0.086	0.099	0.117	0.155	0.187
High Earnings/High Volatility	425	0.123	0.069	0.093	0.117	0.149	0.188
Difference		***0.006***	***0.017***	***0.006***	***0.000***	***0.006***	***-0.001***
Realized earnings for t+1							
Portfolios	N	Mean	10%	25%	50%	75%	90%
High Earnings/Low Volatility	424	0.111	0.072	0.084	0.103	0.135	0.168
High Earnings/High Volatility	425	0.091	0.027	0.064	0.095	0.125	0.161
Difference		***0.020***	***0.045***	***0.020***	***0.008***	***0.010***	***0.007***
P-value on tests of the difference between the forecast and realized earnings differences		**0.001**			**0.015**		

The two portfolios are constructed in the following way. First, the full sample is sorted into 20 portfolios on the level of realized earnings at time t, which is defined as the realized earnings for year t as reported in I/B/E/S. Earnings volatility of realized earnings is the standard deviation of realized earnings in the most recent five years as reported in I/B/E/S. Within each earnings portfolio, quintiles of earnings volatility are formed. The high earnings high volatility sub-sample includes observations from the intersection of the highest earnings volatility quintile and earnings level portfolios 17 to 20. The high earnings low volatility sub-sample includes observations from the intersection of the lowest earnings volatility quintile and earnings level portfolios 17 to 20. Portfolio 1 includes the high earnings high volatility sub-sample. Portfolio 2 includes the high earnings low volatility sub-sample. Realized earnings for t +1 is the realized earnings for year t +1 as reported in I/B/E/S. Analyst forecast for t +1 is from I/B/E/S and is defined as the median earnings forecast for t +1 made in the first month after the announcement of realized earnings at time t.

Table 8
Persistence of analyst forecast errors conditional on earnings volatility

Panel A: Descriptive statistics for forecast errors at t and t+1

Variables	N	Mean	Std	Median	Min	Max
FE_t						
Full sample	7,290	-0.0001	0.0124	0.0002	-0.1964	0.4079
Firms with low vol. of Earn.	1,452	0.0000	0.0033	0.0001	-0.0336	0.0287
Firms with high vol. of Earn.	1,452	-0.0021	0.0118	-0.0001	-0.1524	0.0744
FE_{t+1}						
Full sample	7,290	-0.0065	0.0263	-0.0020	-0.2538	0.1773
Firms with low vol. of Earn.	1,454	0.0000	0.0242	0.0006	-0.1964	0.4079
Firms with high vol. of Earn.	1,454	-0.0116	0.0389	-0.0068	-0.2538	0.1773

Panel B: Regression results from the model:

$$FE_{t+1} = b_1 + b_2{*}High\_vol_t + b_3{*}FE_t + b_4{*}High\_vol_t{*}FE_t + \varepsilon_t$$

	Dependent variable = FE_{t+1}	
Explanatory variables	**Predicted sign**	**Estimates (p-value)**
Intercept	(-)	-0.002 *(0.001)*
$High\_vol_t$	(?)	-0.010 *(0.000)*
FE_t	(+)	0.741 *(0.001)*
$FE_t{*}High\_vol_t$	**(-)**	-0.571 (0.013)
N	2,906	
Adjusted R-squared	3.90%	

The full sample includes all analyst forecasts from the summary IBES tape from year 1984 to year 2004 with the additional requirement that data on realized earnings volatility for the most recent five years is available. After deleting the top and bottom 1 percent of forecast errors, the final sample consists of 7,290 observations. $High\_vol_t$ is an indicator variable, which is coded as 1 if a firm is in the top quintile of earnings volatility and 0 if a firm is in the bottom quintile of earnings volatility. FE_t is defined as the realized year t earnings minus the last median analyst forecast for year t prior to the announcement of year t earnings. FE_{t+1} is defined as realized earnings for t +1 minus the first median analyst forecast for t +1 made immediately after the announcement of year t earnings. The regression only includes firm-year observations in the lowest and highest earnings volatility quintiles.